Tiernan O'Kane

Portfolio: tiernanokane.com | linkedin.com/in/tiernan-okane | tiernan.okane@gmail.com | 774-804-0021

Mechanical engineer who enjoys solving practical problems across manufacturing, design, and analysis.

Engineering Experience

Build The Future Rotational Engineer - Prysmian (Encore Wire) *Summer 2026 – Present*

- Implemented the plant's first live production tracking system, connecting 28 devices across 12 machines to the network and building an Ignition SCADA dashboard for live machine data and KPIs
- Developing a system to track inventory and quality test results in the new medium voltage plant, including the SQL test database structure and vendor software integration to eliminate manual data entry
- Leading a project to upgrade the fume extraction machines, meeting with vendors and stakeholders to set direction, designing a new filter holder, and collecting performance data for a cost-benefit analysis of component upgrades
- Investigating the root cause of sparks that generate scrap, using Minitab to analyze two months of data and pinpoint where sparks originate, and now running trials on how machine settings affect spark count
- Performed root cause analysis on footage discrepancies across machines and implemented corrective action
- Designed machine tooling upgrades to run new products and alert operators to damaged product
- Cataloged wire break samples with a team to reduce preventable breaks caused by bad material from the rod mill

Intern / Part-Time Engineer - Covestro *Summer 2025 – Spring 2026*

- Managed 9 cross-functional engineering projects across process analysis, safety design, and SOP standardization
- Conducted 300+ lab moisture tests, reducing total test time by 40% and improving process consistency
- Designed and fabricated 4 safety guards and modeled 4 trim winder concepts to enhance safety and minimize downtime
- Collected baseline data on polycarbonate inclusion counts, identifying opportunities for quality improvement

Analysis Lead - Senior Design (1st Place) *Fall 2025 – Spring 2026*

- Led mechanical analysis for an assembly line learning tool, achieving above a 95% build success rate
- Verified structural integrity with MATLAB and Ansys Mechanical, confirming minimum FOS with hand calculations
- Tested block placement forces over 10,000 cycles to model material decay in 3D-printed ABS

Project Manager - Engineers Without Borders *Fall 2023 – Spring 2026*

- Managed design, site evaluation, and CAD/BOM documentation for a rainwater catchment system and greenhouse restoration

Double Pendulum - Independent Project *Fall 2024 – Spring 2025*

- Designed and built a modular double pendulum using DFMA, CNC, waterjet, 3D printing, and Arduino

Skills

Software: SolidWorks, Creo Parametric, Fusion 360, Inventor, Ansys Mechanical, MATLAB, Minitab, Ignition SCADA, SQL
Process & Technical: Root Cause Analysis, DFMA, SOP Development, KPI Dashboards, 3D Printing, Arduino, Spanish

Education & Certifications

University of Massachusetts Amherst *2022 – 2026*
B.S. Mechanical Engineering, *cum laude* Major GPA: 3.9
Study Abroad: University College Cork, Ireland (Spring 2024) | University of Minnesota (Fall 2023)

Certifications: Ignition 8.3, Inductive Automation (2026) | Google Project Management (in progress)

Leadership & Activities

Flotilla Skipper - Sail Caribbean, British Virgin Islands *Summer 2024*

- Skippered 46-foot catamarans, leading crews of 10–11 students and managing safety and operations

Senior Sailing Instructor - Hyannis Port Yacht Club *Summers 2021 – 2023*

- Directed curriculum design and implemented safety and leadership initiatives for youth sailors

Activities: Pi Tau Sigma, UMass Fencing Team, UMass Sailing Team - Captain, Sailed Atlantic Crossing